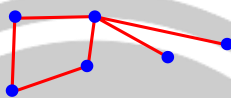


1.2 Connectivity

Name	Repeated Vertex (Vertices)	Repeated Edge(s)	Open	Closed
Walk (open)	Yes	Yes	Yes	
Walk (closed)	Yes	Yes		Yes
Trail	Yes	No	Yes	
Circuit	Yes	No		Yes
Path	No	No	Yes	
Cycle	No	No		Yes

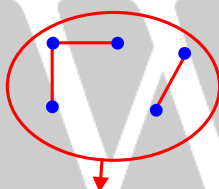
1.2.1 Connected graph:

For every two pair of vertices, there must exist a path between them.

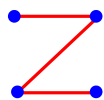


1.2.2 Disconnected graph:

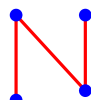
If we can find at least one pair of vertices, such that there is no path available b/w these 2 vertices, then the graph is said to be disconnected graph.



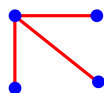
- If G is connected then $\bar{G}$ may be connected or disconnected


 G

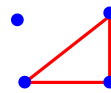
Connected


 $\bar{G}$

Connected


 G

Connected

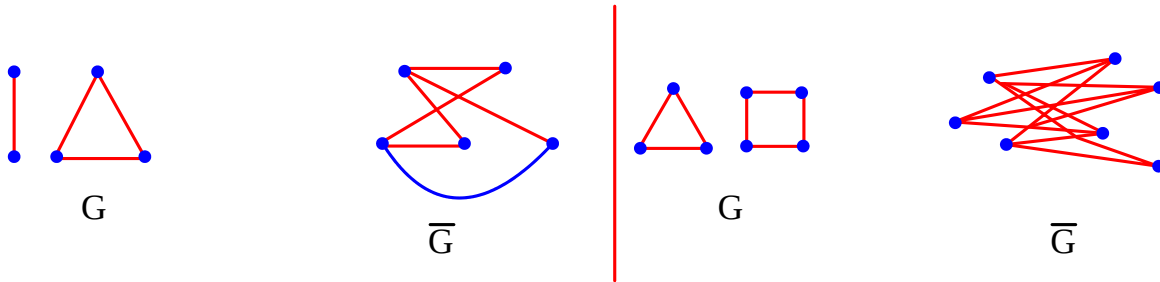

 $\bar{G}$

disconnected

1.2.3 Theorem 6:

If G is disconnected, then $\bar{G}$ is connected.

Example:



1.2.4 Range of edges for a connected graph (k=1):

(k is connected components)

Tree

- Minimum no of edges required to get a possibility to make graph connected with in vertices is $n-1$.

$$(n-1)_{\text{Tree}} \leq e \leq \frac{n(n-1)}{2}$$

- The connected graph with $n-1$ edges is doesn't have a cycle.
- This graph is known as minimally connected graph.
- In this kind of graph, there will be a unique path b/w any two pair of vertices.
- This kind of graph is called a tree (a connected graph with no cycles)

1.2.5 Range of edges for a disconnected graph:

- Edges range b/w

$$n-k \leq e \leq \frac{(n-k)(n-k+1)}{2}$$

Proof:

Here let's say we have k components with $n_1, n_2, \dots, n_k$ components

$$\therefore n_1 + n_2 + \dots + n_k = n$$

For min no. of edge, each compared must be minimally connected.

$\therefore$ min no of edges is

$$n_1 - 1 + n_2 - 1 + \dots + n_k - 1$$

$$= (n_1 + n_2 + \dots + n_k) - k = n - k$$

Note:

1. Let G be a graph of order n . If

$$\deg u + \deg v \geq n - 1$$

nonadjacent vertices u and v of G , then G is connected and $\text{diam}(G) \leq 2$.

2. If G is a graph of order n with $\delta(G) \geq (n-1)/2$, then G is connected.

3. A directed graph is strong connected if there is a path from a to b and from b to a whenever a and b are vertices in the graph.

4. A directed graph is weakly connected if there is a path between every two vertices in the underlying undirected graph.

$$5. k(G) \leq \lambda(G) \leq \min_{v \in V} \deg(v).$$

6. Simple graph G with n vertices is connected if it has more than $(n - 1)(n - 2)/2$ edges.
7. Let $G = (V, E)$ be a loop-free graph with $n (\geq 2)$ vertices. If $\deg(v) \geq (n - 1)/2$ for all $v \in V$, then G has a Hamilton path.
8. If $G = (V, E)$ is a loop-free undirected graph with $|V| = n \geq 3$, and if $|E| \geq \binom{n-1}{2} + 2$, then G has a Hamilton cycle.
9. If $G = (V, E)$ is a loop-free undirected graph with $|V| = n \geq 3$, and $\deg(v) \geq n/2$ for all $v \in V$, then G has a Hamilton cycle.
- (10) If G_1, G_2 are (loop-free) undirected graphs, G_1, G_2 are isomorphic if and only if $\bar{G}_1, \bar{G}_2$ are isomorphic.
- (11) If G is an undirected graph or multigraph with no isolated vertices, then we can construct an Euler trail in G if and only if G is connected and has exactly two vertices of odd degree.
- (12) $k(G) \leq \lambda(G) \leq \delta(G) \leq \Delta(G) \leq n - 1$

1.3 Planarity

A graph (or multigraph) G is called planar if G can be drawn in the plane with its edges intersecting only at vertices of G . Such a drawing of G is called an embedding of G in the plane.

Kuratowski's Theorem. A graph is nonplanar if and only if it contains a subgraph that is homomorphic to either K_5 or $K_{3,3}$.

Let $G = (V, E)$ be a connected planar graph or multigraph with $|V| = v$ and $|E| = e$. Let r be the number of regions in the plane determined by a planar embedding (or, depiction) of G ; one of these regions has infinite area and is called the infinite region. Then $v - e + r = 2$.

Let $G = (V, E)$ be a loop-free connected planar graph with $|V| = v, |E| = e > 2$, and r regions. Then $3r \leq 2e$ and $e \leq 3v - 6$.

1.3.1 Coloring:

Note: Every MIS will always be MDS. But reverse need not to be true

Domination number $\leq$ Independence number.

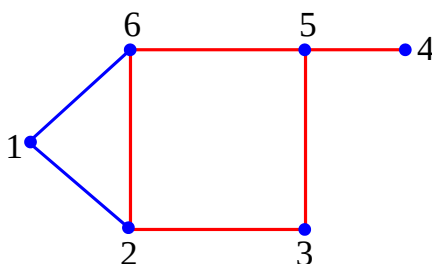
$$\alpha(G) \leq \beta(G)$$

1.3.2 Theorem 7 :

Sum of size of minimum vertex cover and size of maximum independent set is equal to number of vertices

1.4 Covering:

It is set of edges such that all vertices should incident on at least one edge.



$\{16, 12, 65, 53, 54\}$
 $\{16, 54, 23\}, \{16, 12, 53, 54\}$
 $\{12, 16, 65, 23, 53, 62, 54\}$

Set of all edge is also a covering set

- It is also known as edge covering set.

1.4.1 Minimal Covering set:

It is a covering set from which we can't remove new elements(edge).

$\{16, 12, 53, 54\}$. $\{16, 54, 23\}$ are MCS

1.4.2 Covering Number (C(G)):

It is no of edges present smallest covering set.

For above graph $C(G) = 3$

1.5 Perfect Matching:

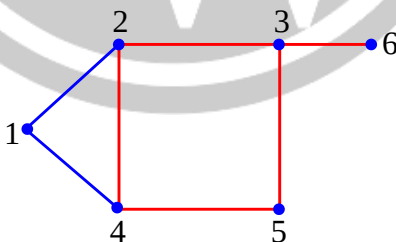
A matching is said to be perfect matching if every vertex in the graph is matched (or)

Induced degree of all the vertices is 1.

1.5.1 Induced degree:

The degree of a vertex in a matching is called indeed degree

Example:



$\{12, 45, 36\}$ is perfect matching

Note: Every perfect matching is maximal, but reverse need not to be true.

If perfect matching exists, then no. of vertices will always be even but reverse need not to be true.

A graph may contain more than one perfect matching.

Total no. of perfect matching possible for a complete graph with $2n$ vertices is $\frac{(2n)!}{2^n \cdot n!}$

